Task 1

//Run the methodA() and methodB() on an Instance of Test two times and explain the answer.

```
public class Test
{
int sum;
public int y;

public void methodA()
{
int x=0, y =0;
y = y + 7;
x = y + 11;
sum = x + y;
System.out.println(x + " " + y+ " " + sum);
}

public void methodB()
{
int x = 0;
y = y + 11;
x = x + 33 + y;
sum = sum + x + y;
System.out.println(x + " " + y+ " " + sum);
}
}
```

```
public class TestDrvier
{
public static void  main(String args[])
{
Test t1=new Test();
t1.methodA();
t1.methodB();
t1.methodA();
t1.methodB();
}
}
```

	sum	y	MethodA		MethodB	Output
			x	y	x	
	0	0				
t1.methodA();			0	0		
	25		18	7		18 7 25
t1.methodB();		11			0	
	80				44	44 11 80
t1.methodA();			0	0		
	25		18	7		18 7 25
t1.methodB();		22			0	
	102				55	55 22 102

Task 2

```
public class Q3
{
public static void main(String args[])
  {
    String test = "";
int i = 5, j = 0, k = 15;
while (i< 10)
{
k- =1;
 j = k;
          while (j > 10 )
          {
          if (j % 2 == 0)
          {
          test = "<--";
          test =  test + i + 2 + "-->" + (j / 2);
          }
          else
          {
          test = "-->";
          test =  "-->" + (i / 2) + test + j;
          }
          System.out.println(test);
                --j;
              }
i++;
    }
  }
}
```

test	i	j	k	output
“”	5	0	15	
<--		14	14	
<--52-->7		13		<--52-->7
-->				
-->2-->13		12		-->2-->13
<--				
<--52-->6		11		<--52-->6
-->				
-->2-->11	6	10		-->2-->11
-->		13	13	
-->3-->13		12		-->3-->13
<--				
<--62-->6		11		<--62-->6
-->				
-->3-->11	7	10		-->3-->11
…	…	12	12	…

Task 4 //important for pass by value and pass by reference
//Run the methodA() on an Instance of Test4 two times and explain the answer.

```
public class Test4{
public int sum;
public int y;

public void methodA(){
int x=0, y =0;
int [] msg = new int[1];
msg[0] = 5;
   y = y + methodB(msg[0]);
   x = y + methodB(msg, msg[0]);
sum = x + y + msg[0];
System.out.println(x + " " + y+ " " + sum);
  }
```

msg[] 5

```
private int methodB(int mg2[] , int mg1){
int x = 0;
   y = y + mg2[0];
   x = x + 33 + mg1;
sum = sum + x + y;
mg2[0] = y + mg1;
   mg1 = mg1 + x + 2;
System.out.println(x + " " + y+ " " + sum);
return sum;
  }
```

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```
private int methodB(int mg1){
int x = 0;
int y = 0;
   y = y + mg1;
   x = x + 33 + mg1;
sum = sum + x + y;
this.y = mg1 + x + 2;
System.out.println(x + " " + y+ " " + sum);
return y;
  }
}
```

msg[]			
index	value		
0	5	50+5=55	...

sum	y	methodA			methodB(int mg2[], int mg1)			methodB(int mg1)			Output
		x	y	msg[]	x	*mg2[]*	*mg1*	x	y	*mg1*	
0	0	0	0		0		5	0	0	5	
0+38+5= 43	5+38+ 2=45	5+131= 136	0+ 5= 5		0+3 3+5 =38		5+38+2= 45	0+ 33 +5 =3 8	0 + 5 = 5		38 5 43
43+38+50= 131	45+5= 50										38 50 131
136+5+55= 196											136 5 196
cont....											

Task 6

```
/*
What is the output for the following code
sequence?
FinalT3A fT3A = new FinalT3A();
fT3A.methodA();
fT3A.methodB(6,8);
*/

public class FinalT3A{
public int sum;
public int y;

public void methodA(){
int x=0, y =0, j = 0;
while (j < 2){
      y = y + j;
      x = j + methodB(y , j);
sum = x + y;
System.out.println(x + " " + y+ " " + sum);
j++;
    }
  }
public int methodB(int p, int k){
int x = 0;
    y = y + k + 1;
    x = x + 3 - p;
sum = sum + x + y;
System.out.println(x + " " + y+ " " + sum);
return sum;
  }
}
```

sum	y	Method A			Method B			output
		x	y	j	x	*p*	*k*	
0	0	0	0	0	0	0	0	3 1 4
(0+3+1)=4	(0+0+1)=1	(0+4)=4	0	1	(0+3-0)=3	1	1	4 0 4
(4+0)=4	(1+1+1)=3	(1+9)=10	(0+1)=1	2	0	6	8	2 3 9
(4+2+3)=9	(3+8+1)=12	22	2		(0+3-1)=2			10 1 11
(10+1)=11					(0+3-6)= -3			-3 12 20
(11-3+12)=20								